

Christine Nattrass
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- Education**
- Yale University Physics M.Sc. 2005, M.Phil. 2006, Ph.D. candidate - anticipated graduation May 2009
 - Colorado State University B.S. 2003 cum laude, Physics, Biochemistry, and Physical Science majors, Chemistry, Math, and German minors

Goal • Post doctoral position in heavy ion physics

Doctoral Work • Advisor: John Harris. Energy, system size, and particle type dependence of high p_T triggered correlations in STAR at RHIC with an emphasis on the near-side. Analyzed Cu+Cu collisions at $\sqrt{s_{NN}} = 200$ and 62 GeV, and Au+Au collisions at $\sqrt{s_{NN}} = 62$ GeV. Studied correlations of h , K^0_s , Λ , and Ξ particles.

- Awards**
- Gertude Goldhaber Prize 2008
 - AAPT Teaching Assistant Award (June 2008)
 - Certificate of Appreciation from RHIC/AGS User's Executive Committee (May 2008)

- Publications**
- Papers on high p_T triggered correlations of strange particles in Cu+Cu collisions at $\sqrt{s_{NN}} = 200$ GeV and energy and system dependence of high p_T triggered correlations – in preparation
 - “Energy and system dependence of high- p_T triggered two-particle near-side correlations”, C. E. Nattrass (STAR) Hot Quarks Proceedings, [submitted to European Physical Journal C on Sept. 30, 2008](#)
 - “System and energy dependence of strange and non-strange particle correlations in STAR at RHIC”, C. E. Nattrass (STAR) Quark Matter 2008 Proceedings, [J. Phys. G: Nucl. Part. Phys. 35 No 10 \(October 2008\) 104110](#)
 - “System size dependence of strange particle correlations in Cu+Cu and Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV in STAR at RHIC”, C. E. Nattrass (STAR) Strange Quark Matter 2007 Proceedings, [J. Phys. G: Nucl. Part. Phys. 35 No 4 \(April 2008\) 044063](#)
 - Author on [STAR papers](#) submitted after Apr. 28, 2006
 - “Elastic properties of $\text{LaSn}_x\text{Ni}_{5-x}$ compounds” (O. Yeheskel, C. E. Nattrass, R. G. Leisure, I. Jacob, R. C. Bowman), [J Appl Phys 95 \(3\): 1089-1095 Feb. 1 2004](#).
 - “Temperature dependence of the elastic moduli of polycrystalline $\text{LaAl}_x\text{Ni}_{5-x}$ and $\text{LaSn}_x\text{Ni}_{5-x}$ ” (J.E. Atteberry, D.S. Agosta, C.E. Nattrass, R. G. Leisure, I. Jacob, R. C. Bowman) [Journal of Alloys and Compounds, Volume 376, Issues 1-2, 11 Aug. 2004, Pages 139-144](#).

- Presentations**
- [International Conference on Women In Physics](#) – poster presentation (October 2008)
 - [BNL workshop on the Ridge](#) – invited talk (September 2008)
 - [Hot Quarks](#) – invited talk (August 2008)
 - Gordon Conference – invited talk (June 2008)
 - [Quark Matter](#) – contributed talk (February 2008)
 - [Palaver Seminar](#), Frankfurt University (January 2008)
 - [Trojkráľ'ová Konferencia](#) (Slovakia) – contributed talk (January 2008)
 - [American Physical Society Division of Nuclear Physics](#) – contributed talk (October 2007)
 - [Strangeness in Quark Matter](#) – contributed talk (June 2007)
 - [American Physical Society April Meeting](#) – contributed talk (April 2007)
 - [American Physical Society Division of Nuclear Physics](#) – contributed talk (October 2006)
 - Gordon Conference – poster presentation (June 2006)
 - American Physical Society March Meeting – contributed talk (March 2003)
 - American Physical Society Four Corners meeting - contributed talk (Nov. 2002)
 - American Physical Society Four Corners meeting - poster presentation (Nov. 2001)

- Service Work**
- Helped with the calibration of the STAR Silicon Vertex Tracker (SVT) (4 months)
 - Studied V^0 reconstruction with the SVT in the year 5 run for the SVT review (1 month)
 - Helped re-cable SVT and make new electronic boards for the read-out system (1 week)
 - Studied calibration of SVT using the lasers (3 weeks)
 - Quality assurance for the year 7 run for the strangeness working group (2 weeks)
 - One of the computer system administrators for the Yale heavy ion group

- Committee Work**
- Brookhaven National Laboratory (BNL) RHIC/AGS User's Executive Committee student/post-doctoral fellow representative June 2006 – June 2008
 - BNL Association for Students And Post-doctoral fellows board Oct. 2006 – Oct. 2007
 - Juniors' representative on the STAR council since April 2008
- Community Service**
- Part of the American delegation to the 2008 IUPAP International Conference on Women in Physics
 - On organizing committee for Conference for Undergraduate Women in Physics at Yale in 2008 and 2009
 - Volunteer for Girls' Science Initiatives, a science education program for middle school girls (2007-2008)
 - Co-organizer for 2008 RHIC/AGS User's Meeting workshop on diversity in physics
 - Volunteer for BNL Summer Sundays public tours of RHIC and STAR in 2006 and 2007
- Teaching**
- Attended Yale graduate school course “Fundamentals of Teaching in the Natural Sciences”
 - Level II International Tutor Certification from the College Reading and Learning Association
 - Teaching assistant introductory physics lab for biological scientists (Fall 2003, Spring 2004, Fall 2004, Spring 2005, Summer 2005, Fall 2005, Spring 2006, Fall 2006); differential equations (Spring 2007); mathematical methods for physicists (Fall 2007); introductory physics (Spring 2008); Quantum Physics and Beyond seminar (Fall 2008)
 - Supervised Xieyue Fan on her senior thesis completed May 2009
- Undergraduate Research Experience**
- 4-15 hours per week, Jan. 2002 - May 2002, Aug. 2002 - current, resonant ultrasound spectroscopy Supervisor: Dr. Bob Leisure, leisure@lamar.colostate.edu
 - NSF International REU through Northeastern University at CERN, Geneva, Switzerland, 40 hours per week, June 2002 - Aug. 2002, helped in construction of ultrahigh vacuum system Supervisors: Dr. Cody Storry, Dr. Gerald Gabrielse, Cody.Storry@cern.ch, gabrielse@physics.harvard.edu
 - NSF REU at Colorado State University, about 40 hours per week, June 2001 - Aug. 2001, expressing and purifying membrane proteins from E. coli and R. sphaeroides for EPR Supervisor: Dr. Craig Schenck, 970-491-6442, cschenck@lamar.colostate.edu
 - NSF International REU through Colorado State University at Leiden Universiteit, Leiden, the Netherlands in the Huygens Laboratory (Physics department), about 40 hours per week, Jan. 2001 - June 2001, EPR on spin labeled proteins, Supervisor: Dr. Martina Huber, mhuber@rulhml.leidenuniv.nl
 - 10-20 hours per week, Aug. 2000 - Jan. 2001, Aug. 2001 - Jan. 2001, expressing and purifying the light harvesting center from R. sphaeroides, mutagenesis for EPR Supervisor: Dr. Craig Schenck, 970-491-6442, cschenck@lamar.colostate.edu
 - Summer Undergraduate Research Experience, University of North Carolina, Chapel Hill, about 40 hours per week, June 2000 - Aug. 2000, computer modeling of trypsin-like serine proteases Supervisor: Dr. Alex Tropsha, tropsha@email.unc.edu
- Undergraduate Awards**
- Barry Goldwater Scholar (2002-2003 school year)
 - Phi Beta Kappa honors society (Spring 2002- current)
 - Sigma Xi honors society associate membership (Spring 2002 – current)
 - CSU Physics Department Weber Scholar (2002-2003 school year)
 - SPS Outstanding Student Award honorable mention
 - CSU College of Natural Science Dean's List (Fall 1998, 1999, 2001, and Spring 1999, 2000, 2002, 2003)
 - AFCEA's General Wickham Scholarship (2002-2003 school year)
 - CSU Honors Program Distinguished Scholar Scholarship (Fall 1998- Spring 2002)
 - CSU President's Scholarship (Fall 1999 - Spring 2001)
 - CSU College of Natural Sciences Students First Scholarship (Fall 2000 - Spring 2002)
 - Sigma Pi Sigma (Spring 2003)
 - Golden Key National Honors Society (Spring 99-current)
 - SPURS (honors society, Fall 1999 - Spring 2000)